Michigan Tree Assisted Migration

1. Background

Michigan forests provide critical ecosystem functions and services (e.g. carbon storage, air and water filtrations, wildlife habitat, and economic benefits) (MEA 2005) and cover over eight million hectares in the state. However, the future health of Michigan forests is threatened by multiple interacting factors, including projected changes in climate, invasive pests and pathogens, and already low resiliency and compromised function due to decreasing species diversity (Walters et al. 2022).

The climate of the Great Lakes region is changing due to global warming. Annual temperatures in the region have risen 0.7-0.9 °C since the beginning of the 20th century, and a recent study has demonstrated an average 11-day extension in growing season length for sugar maple from 1989-2009 (Handler et al. 2014; Burton unpublished data). Mean annual temperatures are projected to rise an additional 1-5 °C through the end of the century (Wuebbles and Hayhoe 2004, Handler et al. 2014) with precipitation projected to increase in winter and spring, but level off or decrease during the summer (Handler et al. 2014). Higher temperatures and lower growing season precipitation may lead to novel environments for species that are adapted to historic climate conditions. Impacts to forest diversity and function may be especially large given several species are located at the southernmost extent of their natural range (e.g. *Pinus resinosa*), likely limiting their adaptive potential (Aitken et al. 2008). While northerly adapted species are likely to become maladaptive in a warmer, drier future, northern Michigan is likely to become more suitable for more southerly distributed genotypes of both current hardwood and softwood species and species distributed further south (Woodall et al. 2009). However, given rapid shifts in climate, seed dispersal and tree establishment limitations may result in natural migration lagging shifts in climate-site suitability (Zhu et al. 2012), potentially leading to high tree mortality (Breshears et al. 2005, Choat et al. 2012) and subsequent losses in productivity (Brzostek et al. 2014) and diversity, contracting species ranges (Murphy et al. 2010, Handler et al. 2014) and changes in forest composition and structure (Mitchell et al. 2023). Ultimately, forest species must either migrate or adapt to avoid local extirpation (Aitken et al. 2008).

The combination of impacts associated with climate change and other factors (e.g. insects and disease) on forests already compromised by low diversity are concerning, and assisted migration (AM) is a tool managers can use to lessen impacts of climate change on forest function and services (Hewitt et al. 2011). AM can be thought of as a suite of strategies that include the purposeful movement of populations that are pre-adapted to future climates or conditions either within their current range or beyond their current range (Park and Talbot 2018, Palik et al. 2022). AM of carefully chosen species/genotypes to address climate change could provide multiple additional related benefits, including a general increase in species/genetic diversity (therefore increased resilience) of forest stands and an increase in the abundance of other important ecological functions.

In order for AM to be a useful tool for forest managers, several considerations must first be addressed; including: 1) selecting enough individuals to maintain genetic diversity within the introduced

populations to avoid inbreeding depression; 2) selecting individuals that match the climates and competitive environments they are being moved to (e.g. cold hardiness, drought tolerance, leaf phenology, growth rate (i.e. competitive ability); and when possible, 3) testing community species assemblages and not only single species. Ultimately, understanding and monitoring the genetic and adaptive potential of forests to cope with rapid climate change will require the development of operational approaches to allow selection and monitoring of forest populations adapted to future climates (Hoban et al. 2022).

To that end, PI Johnson will adapt the resistance, resilience, transition framework (Millar et al. 2007, Nagel et al. 2017, Palik et al. 2022) used in sylviculture to explore assisted migration, adaptation, and the underlying genetics of Michigan forests. In brief, the resistance group include assemblages of trees found in the local forest with some trees better adapted to future climates, the resilience group includes assemblages of trees that are of the same species groups found in the local forest but selected from populations further south and potentially pre-adapted to future climates, and lastly the transition group which includes novel species that are naive to the geographic location but show potential to be adapted to the changing climate and fill a functional role.

The project will be planned and implanted with engagement with a diversity of potential partners, including Federal and other state agencies, tribal nations, conservancies and other non-governmental organizations (NGOs).

2. Scope

We will establish 9 common garden test sites (Table 1) across three climate regions in Michigan (Figure 1) to evaluate the potential for AM of multiple populations and assemblages of seven species planted across three different site conditions (xeric, dry mesic, and mesic) (Table 2). American chestnut plantings will be at the MSU Sand Hill, another common garden sites, or at other yet to be determined Contractor or Michigan DNR (State) provided locations.

Candidate species have been chosen to represent 1) a suite of conifer species at the northern edge of their current distribution that could fill potentially compromised niches/ecological functions created by climate change, 2) more southerly-distributed genotypes of important MI conifer species, 3) naive conifer species that do not currently occur within the site/site quality continuum but could fill a functional role, and 4) American chestnut, a species formerly on the landscape but functionally lost due to the introduction of non-native invasive pathogens.

Table 1: Identified common garden sites across three ecoregions and three site types. Ecoregions include sites within the western upper peninsula (WUP), northern lower peninsula (NLP), and southern lower peninsula (SLP).

	Site Condition		
Region	**Xeric**	**Dry-Mesic**	**Mesic**
WUP	Gwinn C 230	Gwinn C 237	Gwinn C 096
NLP	Grayling C 221	Near Old Higgins Nursery	Old Hardwood Nursery
SLP	Allegan SGA	MSU Clarksville	MSU Russ Forest

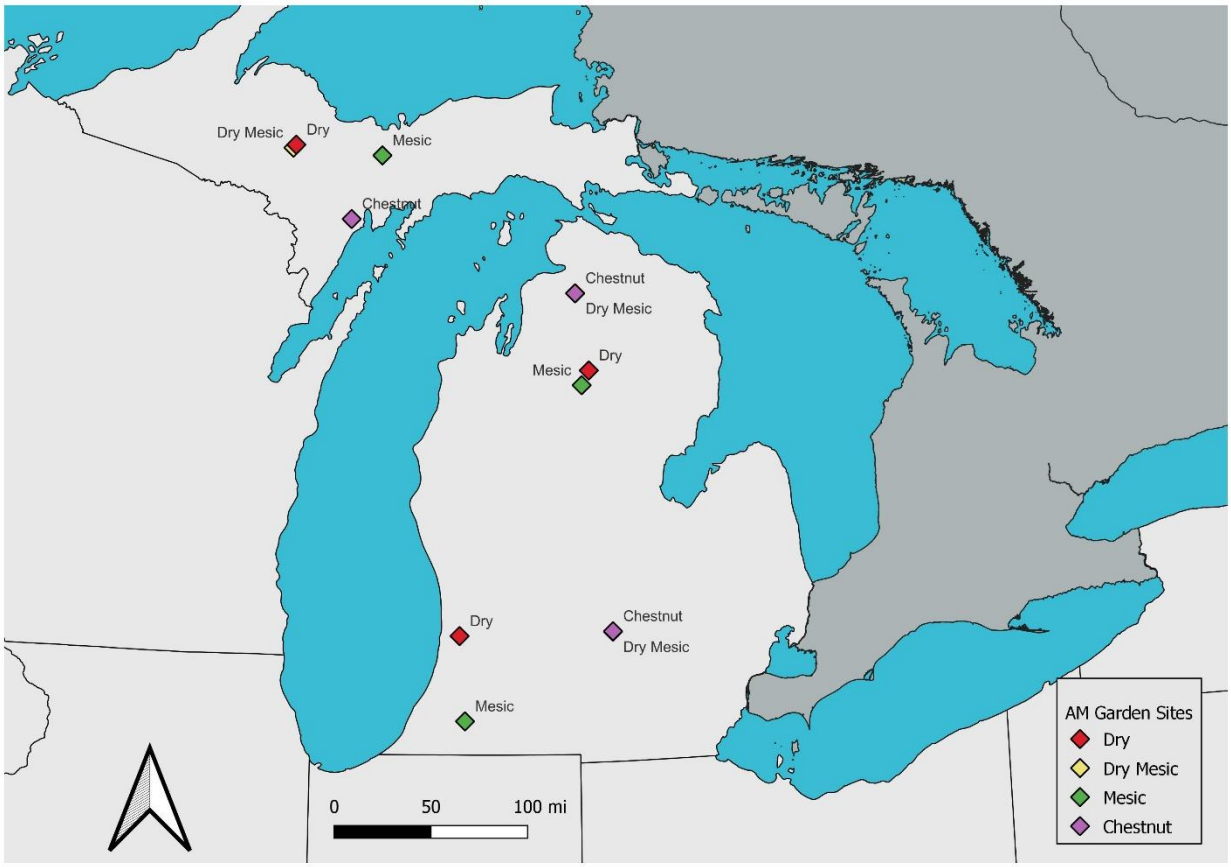

Figure 1: Garden locations across three different climate regions in Michigan. Each climate region is split into three sites that include mesic, dry mesic, and xeric site conditions and will follow the resistance, resilience, transition framework.

Table 2: Selected species to be planted across site types and their associated resistance, resilience, transition categories.

Xeric	category	**Dry-mesic**	category	**Mesic**	category
red pine	resistence	***red pine***	resistance	***red pine***	transition
shortleaf pine	transition	***shortleaf pine***	transition	***shortleaf pine***	transition
white pine	resistence	***white pine***	resistance	***white pine***	resilience
jack pine	resistence	***jack pine***	resilience	**American chestnut**	transition
Virginia pine	transition	***Virginia pine***	transition		
pitch pine	transition	***pitch pine***	transition		
		American chestnut	transition		

3. Approach

3.1. Site Selection

PI Johnson will investigate the effects of climate and soil type on the growth and survival of seven species of trees using a common garden design. The design includes three climate levels (two lower peninsula, one upper peninsula) and three site qualities (xeric, dry mesic, mesic) and seven tree species levels with the response variables associated with survival and growth. The site selections represent different temperature and precipitation (snow zone) regimes allowing simulation for of future climate potentials for different species and species assemblages following the resistance, resilience, and transition framework.

3.2. Seed/Tissue Collection

50 families from each species will be selected in collaboration with a diversity of Federal, State, and NGO partners from across the seven selected species seed zones corresponding to the direction of the best climate analog projections for the region at the end of the century (Hayhoe et al. 2010). The family selections will be informed by collaboration with project partners using the United States Forest Service Eastern Seed Zone Map (Pike et al. 2020) and the USDA Climate Adaptation Tree Atlas (Peters et al. 2020) using the averaged GCM models, with a goal of obtaining a minimum of three families per seed zone. Seed and foliage collection will be accomplished through project personnel designated to manage seed collection, project partners and seed vendors where needed.

Leaf and needle tissue will be collected at the same time as seed/cone collection in FY25 and FY26. GPS coordinates of seed/tissue source trees will be collected along with site characteristics (slope, aspect, disturbance, % cover etc.). Initially, tissue will be collected and placed in bags containing ~10g silica gel desiccant in the field. In the lab, tissue will be stored in a -80 c freezer to await DNA extraction at MSU. DNA will be extracted using Qiagen DNeasy plant kits following manufacture's recommendations. Purified DNA will be stored in -80 c freezers in the MSU Forest Genetics lab until library preparation and sequencing is possible.

3.3. Planting and Data Collection

PI Johnson will sow seeds in heated greenhouses in D33L Deepot containers (2.7" x 8") with D25T racks at the MSU Tree Research Center. Species will be sown over the span of two years, depending on seed availability, with three to four species sown each in FY26 and FY27. The seedlings will be grown for up to two growing seasons in the greenhouse prior to being transplanted into the common garden trials beginning in FY28.

Trials will be planted as randomized complete blocks (ten blocks per trial) using family two tree plots (20 seedlings per family per site). Trees will be planted on 8x8 feet spacing within 2-acre blocks for each tree species (~25 acre/ sites). Each block will include border trees. Each site will be prepared by having stumps removed, weeds controlled, and fencing installed (where not already in place) for American chestnut plantings to reduce herbivory and damage. Sites will be fertilized uniformly to ensure standardized growing conditions.

3.4. Assessment and Analysis

Upon a future change notice to this contract, PI Johnson will conduct assessments of the trials to include measurements of height, diameter, phenology, and survival in the 1st, 3rd, and 5th years, and then every five years after for the duration of the trials (approximately 20 years). Candidate tree performance data and site climate data will be analyzed using appropriate statistical models in the context of current and predicted future climate data to determine matching of best performing species/families with sites/regions for planting. For example, a three-way analysis of variance will be used to test the effects of climate, site, and tree species on the growth and survival of trees. A Tukey's HSD post-hoc test will be used to compare the mean growth and survival of trees for each combination of factor levels. The results of the analysis will show which tree species are best suited for each combination of climate and site types and will provide insight into how future climates and soil will alter forest composition in the future.

4. Monitoring

Future monitoring of the established common garden test sites in Michigan will be accomplished collaboratively by PI Johnson and the State through the Michigan Cooperative Tree Improvement Program over a 20-year timeframe.

5. References

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